ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** ask a question	It begins with a question.
2	**(b)** cell	All living things are made of cells.
3	**(b)** well-being of body, mind and society	WHO definition.
4	**(d)** two terminals	Positive and negative.
5	**(a)** variable	Variables can change.
6	**(a)** tissue	Tissue.
7	**(b)** cold	Droplets carry viruses.
8	**(a)** open circuit	No current.
9	**(a)** number and unit	e.g. 5 m.
10	**(b)** cell, tissue, organ	Small to large.
11	**(b)** all nutrients in right amounts	Balanced diet.
12	**(b)** switch off the supply before fixing a plug	Safe.
13	**(d)** thermometer	Thermometer measures temperature.
14	**(b)** cork	Hooke observed cork.
15	**(a)** cholera	It spreads through water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A quantity or factor that can change (2).	2
17	Smallest structural and functional unit of life (2).	2
18	Complete physical, mental and social well-being (2).	2
19	Cell and wires (1); Switch and bulb (1).	1+1
20	A testable guess or explanation (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Question and hypothesis (1); Test, observe and record (1); Analyse and conclude (1).	1+1+1
22	Cell wall is present in plant cell only (1); Chloroplasts are in plant cells only (1); Plant cell has a large vacuole (1).	1+1+1
23	Clean hands (1); Safe water (1); Vaccination (1).	1+1+1
24	Cell connected to bulb by wires (1); Switch in series (1); Closed switch makes bulb glow (1).	1+1+1
	OR: Length of a pencil 15 cm (1); Mass of a book 500 g (1); Temperature 30 degrees Celsius (1).	1+1+1
25	Human (1); Mango tree (1); Earthworm (1).	1+1+1
	OR: Bathing (1); Brushing teeth (1); Washing hands (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Question: why does bread rise? (1); Hypothesis: yeast makes gas (1); Experiment: bread with and without yeast (1); Observe and record (1); Conclusion (1).	1+1+1+1+1
	OR: Cell membrane: outer covering (1); Cytoplasm: jelly-like (1); Nucleus: control (1); No cell wall (1); No chloroplast (1).	1+1+1+1+1
27	WHO definition (1); Physical health (1); Mental health (1); Social health (1); Examples (1).	1+1+1+1+1
	OR: Cell (1); Wires (1); Switch (1); Bulb (1); Closed circuit allows current (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** positive terminal	Positive.
2	**(d)** one	Only one.
3	**(b)** organ	Organ.
4	**(a)** bacteria	Salmonella.
5	**(d)** closed circuit	Current flows.
6	**(c)** quantitative observation	It contains a number.
7	**(b)** rectangular	Due to the cell wall.
8	**(d)** playing and sleeping well	Healthy habits.
9	**(a)** hot	Heating effect.
10	**(a)** stopwatch	A stopwatch measures time.
11	**(a)** nucleus	The nucleus controls the cell.
12	**(d)** malaria	Anopheles spreads malaria.
13	**(b)** copper	Metal.
14	**(a)** wear goggles while heating	Goggles protect eyes.
15	**(c)** cytoplasm	It fills the cell.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Contains chlorophyll (1); Helps in making food by photosynthesis (1).	1+1
17	Scurvy (1); Anaemia (1).	1+1
18	A safety device with a thin wire (1); Melts when current is excessive (1).	1+1
19	To check reliability (1); To reduce chance errors (1).	1+1
20	Yeast (1); Mushroom (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Don't touch bare wire (1); Don't use wet hands (1); Switch off before repair (1).	1+1+1
22	Only one factor is changed (1); e.g. same size of puri while changing oil temperature (1); So that the result can be explained (1).	1+1+1
23	Functions differ (1); Shape suits function (1); e.g. nerve cell long, RBC round (1).	1+1+1
24	Lemon (1); Amla, orange (1); Prevents scurvy (1).	1+1+1
	OR: High resistance (1); More heat (1); High melting point (1).	1+1+1
25	Metre scale: length (1); Balance: mass (1); Thermometer: temperature (1).	1+1+1
	OR: Nucleus: control (1); Cell membrane: protects and regulates movement (1); Cytoplasm: cell activities (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Qualitative: no number (1); Quantitative: with number (1); Qualitative examples: colour, smell, taste (1); Quantitative: mass, time, length (1); Both are useful in science (1).	1+1+1+1+1
	OR: Scrape the cheek gently (1); Place on a slide with water (1); Stain (1); Cover with a cover slip (1); Observe and draw (1).	1+1+1+1+1
27	Scurvy (1); Rickets (1); Anaemia (1); Goitre (1); Night blindness (1).	1+1+1+1+1
	OR: Iron core (1); Coil (1); Armature and hammer (1); Gong (1); Repeating action (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** protozoa	Plasmodium.
2	**(a)** chemical energy into electrical energy	Cell energy.
3	**(a)** qualitative observation	No number is given.
4	**(c)** no chloroplasts	Root and bulb cells are underground.
5	**(c)** Edward Jenner	1796.
6	**(a)** an electromagnet	To lift iron.
7	**(d)** balance	A balance measures mass.
8	**(b)** chloroplast	Gives green colour.
9	**(c)** vaccine	Polio drops.
10	**(c)** rubber	Not a conductor.
11	**(a)** away from people	Hot liquid may splash out.
12	**(a)** all cells	Every cell has one.
13	**(c)** virus	Highly contagious.
14	**(b)** electricity and magnetism	Magnetic effect.
15	**(a)** check and use them	Results are verified.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Bacteria (1); Through contaminated water (1).	1+1
17	Direct contact of live and neutral wires (1); Large current flows (1).	1+1
18	A test in which only one factor is changed (1); All other factors are kept the same (1).	1+1
19	Cell wall (1); Chloroplast (1).	1+1
20	Malaria (1); Dengue (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Common cold (1); Tuberculosis (1); Measles (1).	1+1+1
22	Conductors: copper, iron, aluminium (1); Insulators: rubber, plastic, glass (1); Conductors let current pass (1).	1+1+1
23	Colour of a flower (1); Smell of a substance (1); Taste of a fruit (1).	1+1+1
24	To see cells clearly (1); Stain gives contrast (1); Safranin or iodine (1).	1+1+1
	OR: Prevent disease (1); Protect others in the community (1); Have saved many lives (1).	1+1+1
25	Short circuit (1); Overloading (1); Poor wires (1).	1+1+1
	OR: Date and title of experiment (1); Observations in a table (1); Conclusion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Independent changes by us (1); Dependent is measured (1); Constant stays same (1); Example: plant growth with water (1); Water amount is independent; height is dependent (1).	1+1+1+1+1
	OR: Cell wall (1); Chloroplast (1); Vacuole (1); Shape (1); Nutrition (1).	1+1+1+1+1
27	Anopheles mosquito (1); Plasmodium (1); Nets (1); Clean surroundings (1); Early treatment (1).	1+1+1+1+1
	OR: Make a circuit with a gap (1); Place the material in the gap (1); Bulb glows: conductor (1); Bulb doesn't glow: insulator (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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